

# Ana Lorena Ortiz Loyola

Frankfurt am Main, Germany

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Data science student (Tecnológico de Monterrey, expected 2027) with hands-on Python (Pandas, NumPy) and SQL experience from a data engineering internship at Deutsche Bundesbank (AnaCredit), plus independent research in cryptography and security (LWE, NTRU lattice-based schemes). Reliable, careful, and comfortable working independently, with a genuine interest in applied cybersecurity research.

## Core Competencies

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- **Data Engineering:** Python (Pandas, NumPy), SQL, SQLite, data cleaning and pipeline design
- **Cybersecurity & Cryptography:** Lattice-based schemes (LWE, NTRU), security-parameter and runtime analysis, contributing to a scientific publication
- **Financial Services Experience:** Data quality verification and migration testing at Deutsche Bundesbank (AnaCredit)
- **Programming:** Python, R, SQL, Java, C++
- **Machine Learning:** Scikit-Learn, Keras, TensorFlow, Random Forest, SVM, K-Means
- **Languages & Tools:** German (C1), English (C1), Git

## Professional Experience

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- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**  
*Data Engineer / Intern* *04/2026–07/2026*
  - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to ensure identical business logic after data migration
  - Implemented automated error detection identifying deviations (duplicate IDs, value-range violations) in reported banking data

## Projects

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- **Tecnológico de Monterrey**  
*Lattice-Based Cryptography Research* 10/2024–02/2025
  - Implemented and analyzed the complexity of lattice-based cryptographic algorithms (LWE, NTRU) in Python; contributed to a scientific publication analyzing security parameters and runtime behavior
- **Personal Project (GitHub), built with Claude Code**  
*Banking Data Pipeline – Data Cleaning & Audit Log* 05/2026
  - Built an end-to-end pipeline for simulated bank transactions (150 accounts): normalized SQLite database with foreign-key integrity checks and a fully auditable cleaning log
- **Tecnológico de Monterrey**  
*Predictive Modeling – Healthcare Optimization* 10/2024–12/2024
  - Analyzed 25+ years of national health data; built and validated predictive models (Random Forest, Neural Networks) and K-Means clustering
- **Tecnológico de Monterrey**  
*Air Quality Analysis Mexico City – SVM Classification* 04/2023–06/2023
  - Built SVM models (linear and RBF kernels) to classify critical ozone levels from citywide NO<sub>x</sub>/O<sub>3</sub> data

## Education

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- **Tecnológico de Monterrey** **Mexico**  
*BSc in Data Science* 2022–2027  
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: data analysis & statistics, cryptography & security, machine learning & AI, numerical optimization.
- **Universität Göttingen** **Germany**  
*Exchange Semester* 2025–2026

## Languages

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- German (C1), English (C1, TOEFL iBT 105/120), Spanish (native).

## Publications

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- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

## Honors and Awards

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- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

## References

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- Available upon request.